

ELITE IGCSE ACADEMY

Student Past Paper Solutions

May 2024 P1H Worked Solutions

May 2024 | P1H

31 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 3****TOPIC: Sequences**

May 2024 P1H - May 2024 | P1H

1 Here are the first four terms of an arithmetic sequence.

1 4 7 10

(a) Find an expression, in terms of n , for the n th term of this sequence.

.....
(2)

The n th term of a different arithmetic sequence is $5n + 17$

(b) Find the 12th term of this sequence.

.....
(1)

.....
(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION**Q#: 1****Marks: 3****TOPIC: Sequences**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The sequence is**

The sequence is

$$1, 4, 7, 10$$

2. The common difference is

The common difference is 3.

$$\text{nth term} = 1 + 3(n - 1) = 3n - 2$$

3. Find the nth term

For the other sequence,

$$5n + 17$$

4. The 12th term is

The 12th term is

$$5(12) + 17 = 77$$

FINAL ANSWER $3n - 2$, and 77.

PAST PAPER QUESTION**Q#: 2****Marks: 4****TOPIC: Probability Toolkit**

May 2024 P1H - May 2024 | P1H

- 2 450 students were asked how they travelled to school on Monday.
Each student walked or travelled by bus or travelled by car or travelled by bicycle.
Each student used just one method of travel.

One of these students is chosen at random.

The table shows information about the probability of each method of travel.

Method of travel	walk	bus	car	bicycle
Probability	0.20	x	$2x$	0.26

Work out how many of the 450 students travelled by car.

(Total for Question 2 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 2****Marks: 4****TOPIC: Probability Toolkit**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let the probability of travelling**Let the probability of travelling by bus be x .Then the probability of travelling by car is $2x$.

$$0.20 + x + 2x + 0.26 = 1$$

$$3x = 0.54$$

$$x = 0.18$$

2. Calculate probability

So

$$P(\text{car}) = 2x = 0.36$$

3. Number travelling by car

Number travelling by car:

$$450 \times 0.36 = 162$$

FINAL ANSWER

162.

PAST PAPER QUESTION**Q#: 3****Marks: 2****TOPIC: Prime Factors, HCF & LCM**

May 2024 P1H - May 2024 | P1H

- 3** Find the highest common factor (HCF) of 72 and 108
Show your working clearly.

(Total for Question 3 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 3

Marks: 2

TOPIC: **Prime Factors, HCF & LCM**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Find the LCM

$$72 = 2^3 \times 3^2$$

$$108 = 2^2 \times 3^3$$

2. Find the LCM

For the HCF, take the smaller power of each common prime:

$$\text{HCF} = 2^2 \times 3^2$$

$$2^2 \times 3^2 = 4 \times 9 = 36$$

FINAL ANSWER

36

PAST PAPER QUESTION**Q#: 4****Marks: 3****TOPIC: Percentages**

May 2024 P1H - May 2024 | P1H

- 4 Ava records the number of kilometres she drives each month.

In April, Ava drove 943 kilometres.

This is 15% more than the number of kilometres she drove in March.

Work out the number of kilometres Ava drove in March.

..... kilometres

(Total for Question 4 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 4

Marks: 3

TOPIC: Percentages

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate percentage

April is 15% more than March, so April is 115% of March.

$$1.15 \times \text{March} = 943$$

$$\text{March} = \frac{943}{1.15} = 820$$

FINAL ANSWER

820 kilometres

EA

PAST PAPER QUESTION**Q#: 5****Marks: 4****TOPIC:** Angles in Polygons & Parallel Lines

May 2024 P1H - May 2024 | P1H

- 5 In the diagram, $ABCDE$ is a regular pentagon.

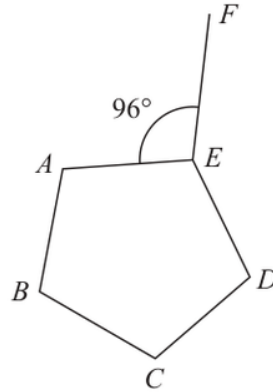


Diagram **NOT**
accurately drawn

Angle $AEF = 96^\circ$

Work out the size of the obtuse angle FED
Show your working clearly.

o

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION**Q#: 5****Marks: 4****TOPIC: Angles in Polygons & Parallel Lines**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The interior angle of a**

The interior angle of a regular pentagon is

$$\frac{(5 - 2) \times 180}{5} = 108^\circ$$

2. Find the gradientAt E , angles around a point add to 360° .

$$\angle FED = 360^\circ - 96^\circ - 108^\circ = 156^\circ$$

FINAL ANSWER 156° .

PAST PAPER QUESTION**Q#: 6****Marks: 5****TOPIC: Solving Linear Equations**

May 2024 P1H - May 2024 | P1H

- 6 (a) Expand and simplify $(m + 5)(m - 8)$

.....
(2)

- (b) Solve $3n - 4 = \frac{5n + 6}{3}$

Show clear algebraic working.

 $n =$
(3)

(Total for Question 6 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 6****Marks: 5****TOPIC: Solving Linear Equations**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$(m + 5)(m - 8) = m^2 - 3m - 40$$

2. (b)

(b)

$$3n - 4 = \frac{5n + 6}{3}$$

$$9n - 12 = 5n + 6$$

$$4n = 18$$

FINAL ANSWER(a) $m^2 - 3m - 40$, (b) $n = \frac{9}{2}$

PAST PAPER QUESTION**Q#: 7****Marks: 5****TOPIC: Set Notation & Venn Diagrams**

May 2024 P1H - May 2024 | P1H

- 7 $\mathcal{E} = \{23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34\}$
 $A = \{\text{even numbers}\}$
 $B = \{23, 29, 31\}$
 $C = \{\text{multiples of 3}\}$

(a) List the members of the set

(i) $B \cup C$

(1)

(ii) $A' \cap C$

(1)

(b) Is it true that $B \cap C = \emptyset$?

Tick (✓) one of the boxes below.

Yes

No

☐☐

Give a reason for your answer.

(1)

The set D has 4 members and is such that $D \cap (A \cup C) = \emptyset$ (c) List the members of set D

(2)

(Total for Question 7 is 5 marks)

PAST PAPER SOLUTION**Q#: 7****Marks: 5****TOPIC: Set Notation & Venn Diagrams**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

$$\mathcal{E} = \{23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34\}$$

$$A = \{24, 26, 28, 30, 32, 34\}$$

$$B = \{23, 29, 31\}$$

$$C = \{24, 27, 30, 33\}$$

2. Calculate value

So

$$B \cup C = \{23, 24, 27, 29, 30, 31, 33\}$$

3. Set of odd numbersAlso A' is the set of odd numbers in $\mathcal{E}$, so

$$A' \cap C = \{27, 33\}$$

4. The statement is true becauseThe statement $B \cap C = \emptyset$ is true because none of 23, 29, 31 is a multiple of 3.**5. Calculate value**For D , we need $D \cap (A \cup C) = \emptyset$. So D must contain the four numbers outside $A \cup C$:

$$D = \{23, 25, 29, 31\}$$

FINAL ANSWER(a)(i) $\{23, 24, 27, 29, 30, 31, 33\}$, (a)(ii) $\{27, 33\}$, (b) Yes, (c) $\{23, 25, 29, 31\}$

PAST PAPER QUESTION**Q#: 8****Marks: 3****TOPIC: Standard & Compound Units**

May 2024 P1H - May 2024 | P1H

- 8 A cylinder is placed on a table.

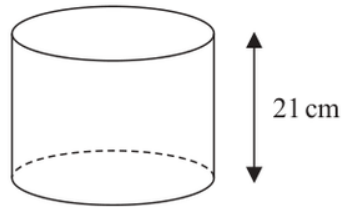


Diagram **NOT**
accurately drawn

The volume of the cylinder is 1575 cm^3

The force exerted by the cylinder on the table is 84 newtons.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the pressure on the table due to the cylinder.

..... newtons/cm²

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION**Q#: 8****Marks: 3****TOPIC: Standard & Compound Units**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The volume of a cylinder**

The volume of a cylinder is

$$\text{volume} = \text{area of base} \times \text{height}$$

2. Calculate area

So the area in contact with the table is

$$1575 \div 21 = 75 \text{ cm}^2$$

3. Use

Use

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$\text{pressure} = \frac{84}{75} = 1.12$$

FINAL ANSWER1.12 newtons/cm².

PAST PAPER QUESTION**Q#: 9****Marks: 3****TOPIC: Powers, Roots & Standard Form**

May 2024 P1H - May 2024 | P1H

- 9 The table gives the amount of rice produced by each of two countries in 2020

Country	Amount of rice (tonnes)
Indonesia	3.5×10^7
Argentina	8.2×10^5

- (a) Write 3.5×10^7 as an ordinary number.

.....
(1)

In 2020, Japan produced 6 780 000 more tonnes of rice than Argentina.

- (b) Work out the amount of rice Japan produced in 2020
Give your answer in standard form.

..... tonnes
(2)

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION

Q#: 9

Marks: 3

TOPIC: **Powers, Roots & Standard Form**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Convert standard form

$$3.5 \times 10^7 = 35000000$$

2. Argentina produced

Argentina produced

$$8.2 \times 10^5 = 820000$$

3. tonnes of rice

tonnes of rice.

4. Japan produced more tonnes than

Japan produced 6780000 more tonnes than Argentina, so

$$820000 + 6780000 = 7600000$$

5. Convert standard form

In standard form,

$$7600000 = 7.6 \times 10^6$$

FINAL ANSWER(a) 35000000, (b) 7.6×10^6 tonnes

PAST PAPER QUESTION**Q#: 10****Marks: 4****TOPIC: Algebraic Roots & Indices**

May 2024 P1H - May 2024 | P1H

10 (a) Simplify $(2p)^0$ where $p > 0$
(1)

$$y^9 \times y^{-3} = y^n$$

(b) Find the value of n $n =$
(1)(c) Simplify fully $(5a^4c^2)^3$
(2)**(Total for Question 10 is 4 marks)**

PAST PAPER SOLUTION

Q#: 10

Marks: 4

TOPIC: **Algebraic Roots & Indices**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

1. (a)

(a) $(2p)^0 = 1$.

2. Use index laws

(b) $y^9 \times y^{-3} = y^{9-3} = y^6$, so $n = 6$.

3. (c)

(c) $(5a^4c^2)^3 = 5^3a^{12}c^6 = 125a^{12}c^6$.

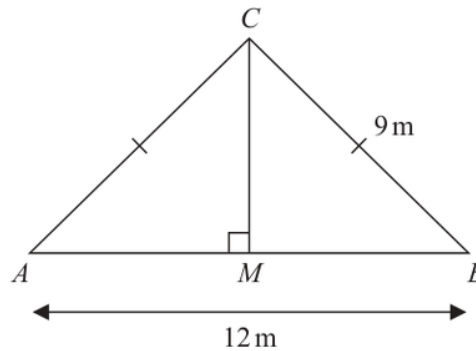
FINAL ANSWER

(a) 1, (b) 6, (c) $125a^{12}c^6$

EA

PAST PAPER QUESTION**Q#: 11****Marks: 4****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

May 2024 P1H - May 2024 | P1H

11 The diagram shows a roof support.Diagram **NOT**
accurately drawn

The roof support is made from four lengths of wood, AB , AC , BC and MC

$$AC = BC = 9\text{ m} \quad AB = 12\text{ m}$$

$$\text{angle } AMC = 90^\circ$$

Lewis is going to buy lengths of wood to make the roof support.

The wood costs 21.50 euros per metre.

Each length of wood he buys has to be a whole number of metres.

Work out the total cost of the wood Lewis needs to buy.

Show your working clearly.

..... euros

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 4****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the gradient**

Since $AC = BC$, the perpendicular support from C meets AB at its midpoint.

$$AM = \frac{12}{2} = 6$$

2. Use trigonometry

In right-angled triangle AMC ,

$$AC^2 = AM^2 + MC^2$$

$$9^2 = 6^2 + MC^2$$

$$MC^2 = 81 - 36 = 45$$

$$MC = \sqrt{45} = 6.708\dots$$

3. A whole number of metres

A whole number of metres must be bought, so buy 7 m for MC .

4. Total length needed

Total length needed:

$$12 + 9 + 9 + 7 = 37$$

5. Cost

Cost:

$$37 \times 21.50 = 795.50$$

FINAL ANSWER

795.50 euros.

PAST PAPER QUESTION**Q#: 12****Marks: 5****TOPIC: Algebraic Fractions**

May 2024 P1H - May 2024 | P1H

12 (a) Factorise fully $6y^2 - 5y - 4$
(2)(b) Express $\frac{2x+1}{4x} + \frac{7-5x}{3x}$ as a single fraction in its simplest form......
(3)

(Total for Question 12 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 12****Marks: 5****TOPIC: Algebraic Fractions**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$6y^2 - 5y - 4 = (3y - 4)(2y + 1)$$

2. (b)

(b)

$$\frac{2x + 1}{4x} + \frac{7 - 5x}{3x}$$

3. Use denominatorUse denominator $12x$:

$$\begin{aligned} & \frac{3(2x + 1) + 4(7 - 5x)}{12x} \\ &= \frac{6x + 3 + 28 - 20x}{12x} = \frac{31 - 14x}{12x} \end{aligned}$$

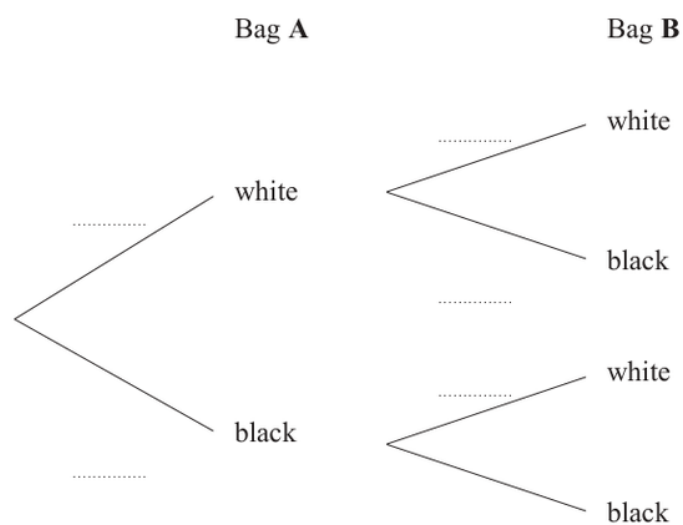
FINAL ANSWER(a) $(3y - 4)(2y + 1)$, (b) $\frac{31 - 14x}{12x}$

PAST PAPER QUESTION**Q#: 13****Marks: 5****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

May 2024 P1H - May 2024 | P1H

13 Harman has two bags of beads.In bag **A**, there are 3 white beads and 7 black beads.In bag **B**, there are 5 white beads and 4 black beads.Harman takes at random a bead from bag **A** and a bead from bag **B**

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Harman takes two beads of the same colour.

(3)

(Total for Question 13 is 5 marks)

PAST PAPER SOLUTION**Q#: 13****Marks: 5****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate probability**

Bag A has 3 white and 7 black counters.

Bag B has 5 white and 4 black counters.

2. The same colour means both

The same colour means both white or both black.

$$\begin{aligned}P(\text{same}) &= \frac{3}{10} \times \frac{5}{9} + \frac{7}{10} \times \frac{4}{9} \\&= \frac{1}{6} + \frac{14}{45} \\&= \frac{43}{90}\end{aligned}$$

FINAL ANSWER

$$\frac{43}{90}$$

PAST PAPER QUESTION**Q#: 14****Marks: 5**TOPIC: **Percentages**

May 2024 P1H - May 2024 | P1H

14 The combined savings of Abel and Bahira are 15 435 dinars.

The savings of Bahira are 45% more than the savings of Abel.

The savings of Bahira are $\frac{3}{2}$ times the savings of Chanda.

Work out the savings of Chanda.

..... dinars

(Total for Question 14 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 14

Marks: 5

TOPIC: Percentages

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Let Abel's savings beLet Abel's savings be A .**2. Calculate percentage**

Bahira has 45% more than Abel, so

$$B = 1.45A$$

3. Their combined savings are

Their combined savings are

$$A + 1.45A = 15435$$

$$2.45A = 15435$$

$$A = 6300$$

4. Calculate percentage

So

$$B = 1.45 \times 6300 = 9135$$

5. Bahira's savings are timesBahira's savings are $\frac{3}{2}$ times Chanda's savings, so

$$9135 = \frac{3}{2}C$$

$$C = 9135 \times \frac{2}{3} = 6090$$

FINAL ANSWER

6090 dinars

PAST PAPER QUESTION**Q#: 15****Marks: 4****TOPIC: Functions**

May 2024 P1H - May 2024 | P1H

15 The function f is defined as

$$f: x \mapsto \frac{3x+1}{x-2}$$

(a) State the value of x that cannot be included in any domain of the function f
(1)(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$ $f^{-1}(x) = \dots$
(3)**(Total for Question 15 is 4 marks)**

PAST PAPER SOLUTION**Q#: 15****Marks: 4****TOPIC: Functions**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Evaluate fraction**

$$f(x) = \frac{3x + 1}{x - 2}$$

2. (a) The denominator cannot be

(a) The denominator cannot be zero, so

$$x - 2 = 0$$

$$x = 2$$

3. (b) Let

(b) Let

$$y = \frac{3x + 1}{x - 2}$$

$$y(x - 2) = 3x + 1$$

$$yx - 2y = 3x + 1$$

$$x(y - 3) = 2y + 1$$

$$x = \frac{2y + 1}{y - 3}$$

4. Evaluate fraction

So

$$f^{-1}(x) = \frac{2x + 1}{x - 3}$$

FINAL ANSWERExcluded value 2, and $f^{-1}(x) = \frac{2x + 1}{x - 3}$.

PAST PAPER QUESTION**Q#: 16****Marks: 4****TOPIC: Combined & Conditional Probability**

May 2024 P1H - May 2024 | P1H

16 There are 20 sweets in a box.

15 of the sweets are red

5 of the sweets are yellow

Fred takes at random 3 sweets from the box.

Work out the probability that Fred takes at least one sweet of each colour from the box.

(Total for Question 16 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 16****Marks: 4****TOPIC: Combined & Conditional Probability**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate probability**

At least one sweet of each colour is the complement of all red or all yellow.

$$P(\text{all red}) = \frac{15}{20} \times \frac{14}{19} \times \frac{13}{18}$$

$$P(\text{all yellow}) = \frac{5}{20} \times \frac{4}{19} \times \frac{3}{18}$$

$$P(\text{at least one of each}) = 1 - P(\text{all red}) - P(\text{all yellow})$$

$$= 1 - \frac{2730}{6840} - \frac{60}{6840}$$

$$= \frac{45}{76}$$

FINAL ANSWER

$$\frac{45}{76}$$

PAST PAPER QUESTION**Q#: 17****Marks: 3****TOPIC: Algebraic Proof**

May 2024 P1H - May 2024 | P1H

17 Show that $\frac{1 + \sqrt{5}}{3 - \sqrt{5}}$ can be written in the form $a + \sqrt{b}$ where a and b are integers.

Show each stage of your working clearly.

(Total for Question 17 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 17****Marks: 3****TOPIC: Algebraic Proof**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Simplify surd**

$$\begin{aligned}\frac{1 + \sqrt{5}}{3 - \sqrt{5}} \times \frac{3 + \sqrt{5}}{3 + \sqrt{5}} &= \frac{(1 + \sqrt{5})(3 + \sqrt{5})}{9 - 5} \\ &= \frac{3 + \sqrt{5} + 3\sqrt{5} + 5}{4} \\ &= \frac{8 + 4\sqrt{5}}{4} = 2 + \sqrt{5}\end{aligned}$$

FINAL ANSWER $2 + \sqrt{5}$, so $a = 2$ and $b = 5$.

PAST PAPER QUESTION**Q#: 18****Marks: 5****TOPIC: Differentiation**

May 2024 P1H - May 2024 | P1H

18 A curve C has equation $y = x^3 - 40x + 1$

Find the coordinates of both the points on C at which the gradient is 8

(..... ,)

(..... ,)

(Total for Question 18 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 18****Marks: 5****TOPIC: Differentiation**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Differentiate**

$$y = x^3 - 40x + 1$$

$$\frac{dy}{dx} = 3x^2 - 40$$

2. Find the gradient

The gradient is 8, so

$$3x^2 - 40 = 8$$

$$3x^2 = 48$$

$$x^2 = 16$$

$$x = 4 \quad \text{or} \quad x = -4$$

3. DifferentiateWhen $x = 4$:

$$y = 4^3 - 40(4) + 1 = -95$$

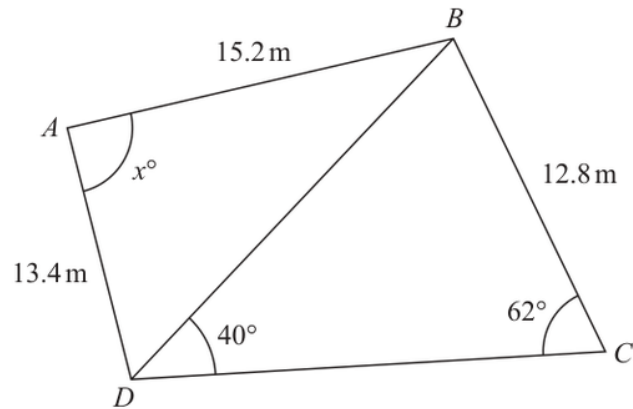
4. DifferentiateWhen $x = -4$:

$$y = (-4)^3 - 40(-4) + 1 = 97$$

FINAL ANSWER $(-4, 97)$ and $(4, -95)$.

PAST PAPER QUESTION**Q#: 19****Marks: 5****TOPIC: Sine, Cosine Rule & Area of Triangles**

May 2024 P1H - May 2024 | P1H

19 Here is quadrilateral $ABCD$ Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to 3 significant figures.

 $x = \dots\dots\dots$ **(Total for Question 19 is 5 marks)**

PAST PAPER SOLUTION**Q#: 19****Marks: 5****TOPIC: Sine, Cosine Rule & Area of Triangles**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**In triangle BCD ,

$$\angle DBC = 180^\circ - 40^\circ - 62^\circ = 78^\circ$$

2. Use trigonometry

By the sine rule,

$$\frac{DB}{\sin 62^\circ} = \frac{12.8}{\sin 40^\circ}$$

$$DB = 17.582 \dots$$

3. Use trigonometryNow use cosine rule in triangle ABD :

$$\cos x = \frac{15.2^2 + 13.4^2 - 17.582 \dots^2}{2(15.2)(13.4)}$$

$$x = 75.577 \dots^\circ$$

FINAL ANSWER **75.6° .**

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Circles, Arcs & Sectors**

May 2024 P1H - May 2024 | P1H

20 The diagram shows a sector OAC of a circle centre O

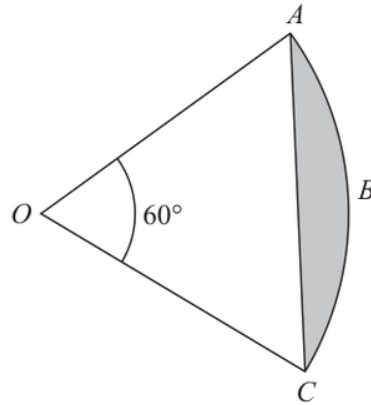


Diagram **NOT**
accurately drawn

Angle $AOC = 60^\circ$

The area of the shaded segment ABC is 38 cm^2

Work out the perimeter of the shaded segment ABC

Give your answer correct to one decimal place.

..... cm

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Circles, Arcs & Sectors**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let the radius of the**Let the radius of the sector be r .**2. The shaded segment is**

The shaded segment is

$$\text{sector area} - \text{triangle area}$$

3. Use trigonometryThe angle is 60° , so

$$38 = \frac{60}{360}\pi r^2 - \frac{1}{2}r^2 \sin 60^\circ$$

$$38 = r^2 \left(\frac{\pi}{6} - \frac{\sqrt{3}}{4} \right)$$

$$r = 20.4815\dots$$

4. The perimeter of the shadedThe perimeter of the shaded segment is the arc AC plus the chord AC .**5. Arc length**

Arc length:

$$\frac{60}{360} \times 2\pi r = \frac{\pi r}{3}$$

6. Chord length

Chord length:

$$2r \sin 30^\circ = r$$

7. Evaluate fraction

So the perimeter is

$$\begin{aligned} & \frac{\pi r}{3} + r \\ &= \frac{\pi(20.4815\dots)}{3} + 20.4815\dots \\ &= 41.9296\dots \end{aligned}$$

FINAL ANSWER

41.9 cm

EA

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Circles, Arcs & Sectors**

May 2024 P1H - May 2024 | P1H

20 The diagram shows a sector OAC of a circle centre O

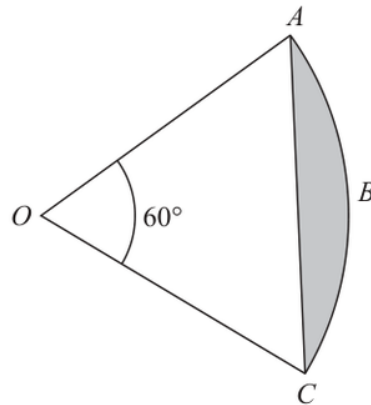


Diagram **NOT**
accurately drawn

Angle $AOC = 60^\circ$

The area of the shaded segment ABC is 38 cm^2

Work out the perimeter of the shaded segment ABC

Give your answer correct to one decimal place.

..... cm

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Circles, Arcs & Sectors**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let the radius of the**Let the radius of the sector be r .**2. The shaded segment is**

The shaded segment is

$$\text{sector area} - \text{triangle area}$$

3. Use trigonometryThe angle is 60° , so

$$38 = \frac{60}{360}\pi r^2 - \frac{1}{2}r^2 \sin 60^\circ$$

$$38 = r^2 \left(\frac{\pi}{6} - \frac{\sqrt{3}}{4} \right)$$

$$r = 20.4815\dots$$

4. The perimeter of the shadedThe perimeter of the shaded segment is the arc AC plus the chord AC .**5. Arc length**

Arc length:

$$\frac{60}{360} \times 2\pi r = \frac{\pi r}{3}$$

6. Chord length

Chord length:

$$2r \sin 30^\circ = r$$

7. Evaluate fraction

So the perimeter is

$$\begin{aligned} & \frac{\pi r}{3} + r \\ &= \frac{\pi(20.4815\dots)}{3} + 20.4815\dots \\ &= 41.9296\dots \end{aligned}$$

FINAL ANSWER

41.9 cm

EA

PAST PAPER QUESTION**Q#: 21****Marks: 2****TOPIC: Transformations of Graphs**

May 2024 P1H - May 2024 | P1H

21 A curve has equation $y = f(x)$

There is one minimum point on this curve.

The coordinates of this minimum point are $(5, -4)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 7)$

(.....,.....)
(1)

(ii) $y = f(x) - 6$

(.....,.....)
(1)

(Total for Question 21 is 2 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 2****TOPIC: Transformations of Graphs**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The minimum point of is**The minimum point of $y = f(x)$ is $(5, -4)$.**2. Calculate value**

For

$$y = f(x + 7)$$

3. the graph moves 7 units

the graph moves 7 units left:

$$(5, -4) \rightarrow (-2, -4)$$

4. Calculate value

For

$$y = f(x) - 6$$

5. the graph moves 6 units

the graph moves 6 units down:

$$(5, -4) \rightarrow (5, -10)$$

FINAL ANSWER $(-2, -4)$ and $(5, -10)$

PAST PAPER QUESTION**Q#: 21****Marks: 2****TOPIC: Transformations of Graphs**

May 2024 P1H - May 2024 | P1H

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(Total for Question 21 is 2 marks)

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May 2024 P1H - May 2024 | P1H

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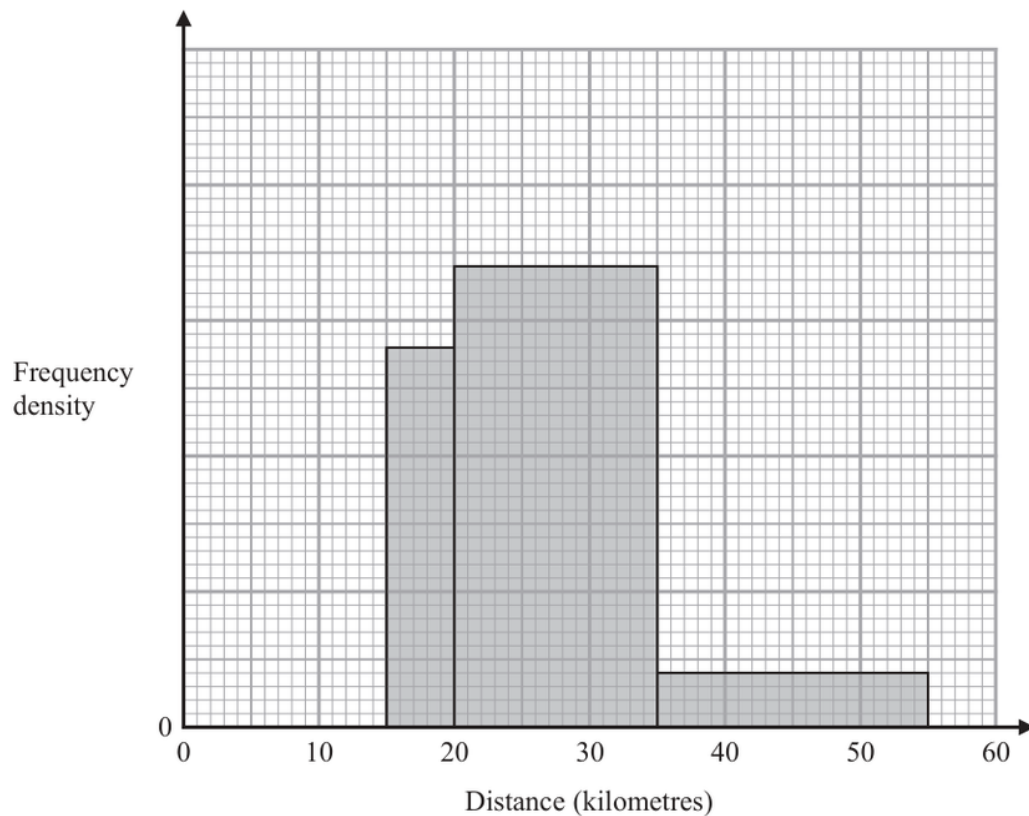
$$(5, -4) \rightarrow (5, -10)$$

FINAL ANSWER $(-2, -4)$ and $(5, -10)$

PAST PAPER QUESTION**Q#: 22****Marks: 3****TOPIC: Histograms**

May 2024 P1H - May 2024 | P1H

- 22** The incomplete histogram shows some information about the distances, in kilometres, that 100 adults ran last week.



All of the adults ran at least 5 kilometres.
 None of the adults ran more than 55 kilometres.
 14 adults ran between 15 kilometres and 20 kilometres.

Complete the histogram.

(Total for Question 22 is 3 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 3****TOPIC: Histograms**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use histogram**

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

2. Use frequency tableFor $15 \leq d < 20$, the frequency is 14 and the class width is 5, so

$$\text{frequency density} = \frac{14}{5} = 2.8$$

3. Use histogram

Using this scale from the histogram:

$$20 \leq d < 35 \quad \text{has frequency density } 3.4$$

$$35 \leq d \leq 55 \quad \text{has frequency density } 0.4$$

4. Calculate value

So the known frequencies are

$$14 + (15)(3.4) + (20)(0.4)$$

$$= 14 + 51 + 8 = 73$$

5. Use frequency table

There are 100 adults in total, so the missing frequency from 5 km to 15 km is

$$100 - 73 = 27$$

6. The missing class width is

The missing class width is 10, so its frequency density is

$$\frac{27}{10} = 2.7$$

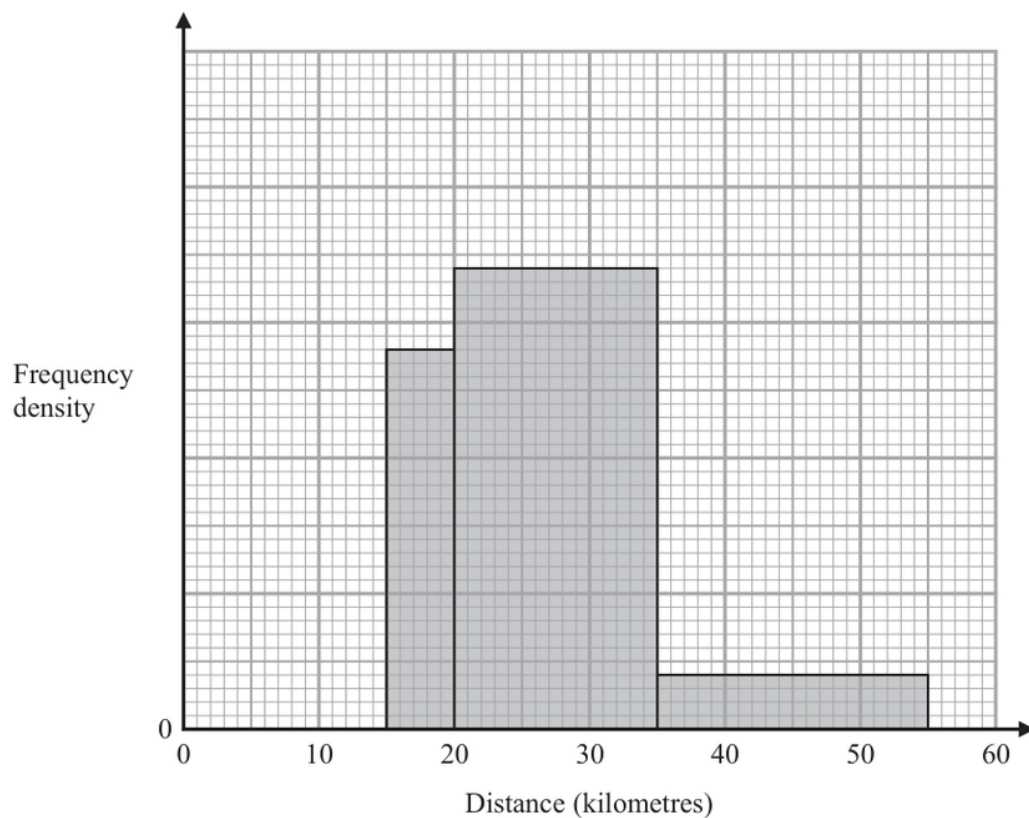
FINAL ANSWER

draw the missing bar from 5 to 15 with frequency density 2.7

PAST PAPER QUESTION**Q#: 22****Marks: 3****TOPIC: Histograms**

May 2024 P1H - May 2024 | P1H

- 22** The incomplete histogram shows some information about the distances, in kilometres, that 100 adults ran last week.



All of the adults ran at least 5 kilometres.
 None of the adults ran more than 55 kilometres.
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(Total for Question 22 is 3 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 3****TOPIC: Histograms**

May 2024 P1H - May 2024 | P1H

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FINAL ANSWER

draw the missing bar from 5 to 15 with frequency density 2.7

PAST PAPER QUESTION**Q#: 23****Marks: 5****TOPIC: Volume & Surface Area**

May 2024 P1H - May 2024 | P1H

- 23** A solid shape is made by removing a hemisphere, shown shaded, from a cone as shown in the diagram.

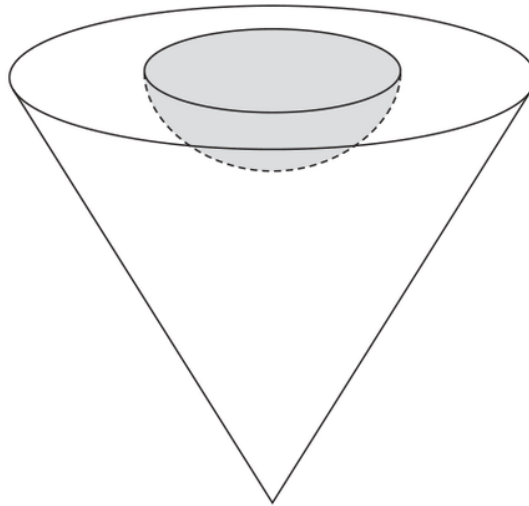


Diagram **NOT**
accurately drawn

The radius of the hemisphere is $2x$ cm

The radius of the base of the cone is $5x$ cm

The vertical height of the cone is $6x$ cm

The volume of the solid shape is 6948π cm³

Work out the **total** surface area of the solid hemisphere that has been removed from the cone.

Give your answer correct to the nearest integer.

..... cm²

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 5****TOPIC: Volume & Surface Area**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Define cone dimensions**

The cone has radius $5x$ and height $6x$.

2. Volume of the cone

Volume of the cone:

$$\begin{aligned}\frac{1}{3}\pi(5x)^2(6x) \\ = 50\pi x^3\end{aligned}$$

3. The removed hemisphere has radius

The removed hemisphere has radius $2x$.

4. Volume of the hemisphere

Volume of the hemisphere:

$$\begin{aligned}\frac{2}{3}\pi(2x)^3 \\ = \frac{16}{3}\pi x^3\end{aligned}$$

5. The remaining solid has volume

The remaining solid has volume 6948π , so

$$\begin{aligned}50\pi x^3 - \frac{16}{3}\pi x^3 &= 6948\pi \\ \frac{134}{3}\pi x^3 &= 6948\pi\end{aligned}$$

6. Cancel

Cancel π :

$$\begin{aligned}\frac{134}{3}x^3 &= 6948 \\ x^3 &= \frac{10422}{67} \\ x &= 5.3781\dots\end{aligned}$$

7. The solid hemisphere removed has

The solid hemisphere removed has radius $2x$.

8. Total surface area of a

Total surface area of a solid hemisphere is

$$3\pi r^2$$

9. Calculate value

So

$$3\pi(2x)^2 = 12\pi x^2$$

$$12\pi(5.3781\dots)^2 = 1090.3903\dots$$

FINAL ANSWER

$$1090 \text{ cm}^2$$

EA

PAST PAPER QUESTION

Q#: 23 Marks: 5

TOPIC: **Volume & Surface Area**

May 2024 P1H - May 2024 | P1H

- 23** A solid shape is made by removing a hemisphere, shown shaded, from a cone as shown in the diagram.

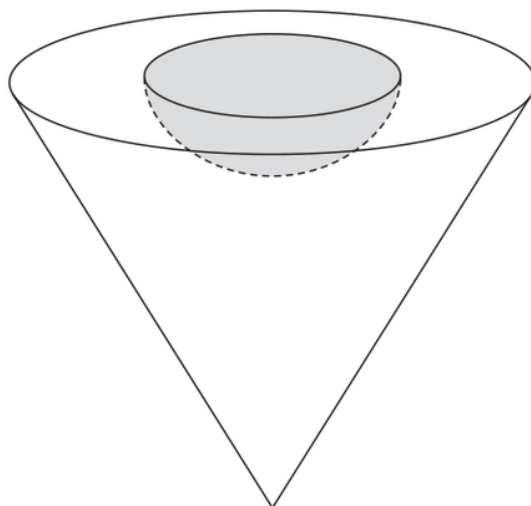


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..... cm²

(Total for Question 23 is 5 marks)

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So

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$$12\pi(5.3781\dots)^2 = 1090.3903\dots$$

FINAL ANSWER

$$1090 \text{ cm}^2$$

EA

PAST PAPER QUESTION**Q#: 24****Marks: 6**TOPIC: **Sequences**

May 2024 P1H - May 2024 | P1H

24 A polygon has n sides, where $n > 5$

The interior angles of the polygon form an arithmetic sequence.

The smallest angle of the polygon is 84°

The common difference of the sequence is 4°

Work out the sum of the interior angles of the polygon.

Show clear algebraic working.

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION**Q#: 24****Marks: 6****TOPIC: Sequences**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The interior angles form an**

The interior angles form an arithmetic sequence with first term 84 and common difference 4.

2. Use trigonometry

So the sum of the angles is

$$\begin{aligned} & \frac{n}{2}(2(84) + (n-1)4) \\ &= \frac{n}{2}(168 + 4n - 4) \\ &= 2n^2 + 82n \end{aligned}$$

3. The sum of the interior

The sum of the interior angles of an n -sided polygon is

$$180(n-2)$$

4. Calculate value

So

$$2n^2 + 82n = 180(n-2)$$

$$2n^2 - 98n + 360 = 0$$

$$n^2 - 49n + 180 = 0$$

$$(n-45)(n-4) = 0$$

5. Solve inequality

Since $n > 5$,

$$n = 45$$

6. Use trigonometry

Therefore the sum of the interior angles is

$$180(45-2) = 7740$$

FINAL ANSWER

7740 degrees

PAST PAPER QUESTION**Q#: 24****Marks: 6****TOPIC: Sequences**

May 2024 P1H - May 2024 | P1H

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Show clear algebraic working.

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(Total for Question 24 is 6 marks)

EAA

PAST PAPER SOLUTION**Q#: 24****Marks: 6****TOPIC: Sequences**

May 2024 P1H - May 2024 | P1H

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FINAL ANSWER

7740 degrees

PAST PAPER QUESTION**Q#: 25****Marks: 4****TOPIC: Completing the Square**

May 2024 P1H - May 2024 | P1H

25 $f(x) = 17 - 3x^2 + 12x$

Write $f(x)$ in the form $a - b(x - c)^2$ where a , b and c are constants.

$f(x) = \dots\dots\dots$

(Total for Question 25 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 25****Marks: 4****TOPIC: Completing the Square**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve inequality**

$$\begin{aligned}f(x) &= 17 - 3x^2 + 12x \\&= -3x^2 + 12x + 17 \\&= -3(x^2 - 4x) + 17 \\&= -3((x - 2)^2 - 4) + 17 \\&= -3(x - 2)^2 + 12 + 17 \\&= 29 - 3(x - 2)^2\end{aligned}$$

FINAL ANSWER

$$f(x) = 29 - 3(x - 2)^2$$

PAST PAPER QUESTION**Q#: 25****Marks: 4****TOPIC: Completing the Square**

May 2024 P1H - May 2024 | P1H

25 $f(x) = 17 - 3x^2 + 12x$

Write $f(x)$ in the form $a - b(x - c)^2$ where a , b and c are constants.

$f(x) = \dots\dots\dots$

(Total for Question 25 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 25****Marks: 4****TOPIC: Completing the Square**

May 2024 P1H - May 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve inequality**

$$\begin{aligned}f(x) &= 17 - 3x^2 + 12x \\&= -3x^2 + 12x + 17 \\&= -3(x^2 - 4x) + 17 \\&= -3((x - 2)^2 - 4) + 17 \\&= -3(x - 2)^2 + 12 + 17 \\&= 29 - 3(x - 2)^2\end{aligned}$$

FINAL ANSWER

$$f(x) = 29 - 3(x - 2)^2$$